# Queues and asynchronous work

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

---

WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Queues and asynchronous work before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Caching and stale information

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Sandboxing and resource limits

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

**Queue**s separate acceptance from completion. Work may be redelivered, so consumers must preserve operation IDs and deduplicate committed effects.

A **queue**d job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.

## Practise with fading help

> Complete the supported Queues and asynchronous work trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Queue**s and asynchronous work. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

---



## Individual reasoning check

> Check the proposed Queues and asynchronous work explanation against the supplied facts before discussing it.

Use the worked case for **Queue**s and asynchronous work. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

---

WORKBOOK · 4 OF 15

## Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

---

WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Queues and asynchronous work facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

---



## W2 Trace the worked case

> Trace the worked case. Apply the supplied Queues and asynchronous work facts and keep your reasoning.

Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

---

## W3 Audit the limit

> Audit the limit. Apply the supplied Queues and asynchronous work facts and keep your reasoning.

A report omits this qualification: **Queue** ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

---



## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Queues and asynchronous work facts and keep your reasoning.

> Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Queues and asynchronous work facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Queues and asynchronous work to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **queue**s and asynchronous work **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **queue**s and asynchronous work.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

---

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

# Pictorial case practice: Queues and asynchronous work

> Queued is not complete. A conversion service queues job J12 rather than processing it during submission. Explain the labelled step before checking the changed condition.

**Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.

States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.

AI ACADEMY / AI-6/22

# Queues and asynchronous work

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Accept request

J12 acknowledges receipt, not completed conversion.



### 2. Start worker

The worker marks J12 running before processing.



### 3. Publish checked result

Only a checked output permits succeeded status.

**CHANGE ONE THING**

The worker crashes before producing checked output.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what the **queue**d J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

---

WORKBOOK · 13 OF 15

## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

---

WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only a checked **output** permits succeeded status.

> 1. Start worker

2. Publish checked result

3. Accept request

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Publish checked result Only a checked **output** permits succeeded status. Asynchronous work separates submission from completion.

Honest status and a job identifier let users track that difference.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The worker crashes before producing checked **output**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The job is failed or awaiting a controlled retry, not succeeded.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The job is failed or awaiting a controlled retry, not succeeded. Receipt is not **evidence** that conversion finished.

**Repair a missing building block**

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version -5 request return, and what permits a later version -5 cache hit?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

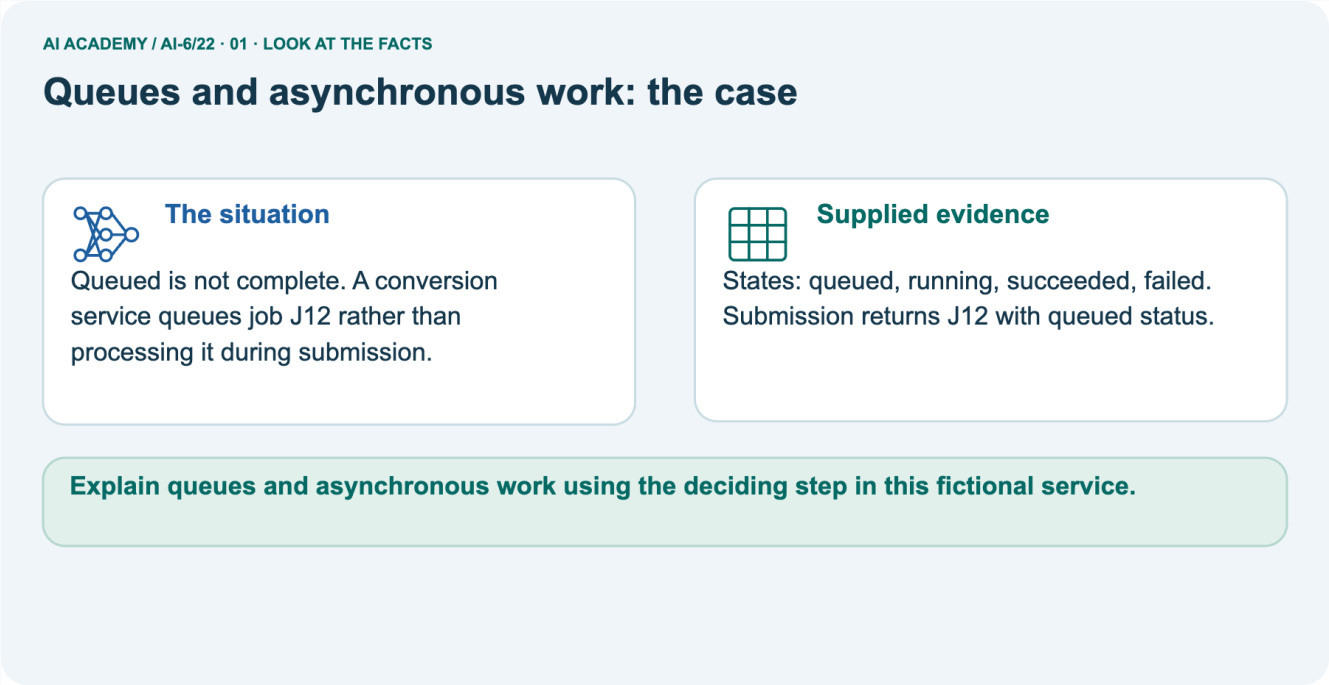
---

WORKBOOK · 15 OF 15

## Visual concept journey

> Explain each Queues and asynchronous work picture from its labels, then record a relevant limit.

## 01. Queues and asynchronous work: the case



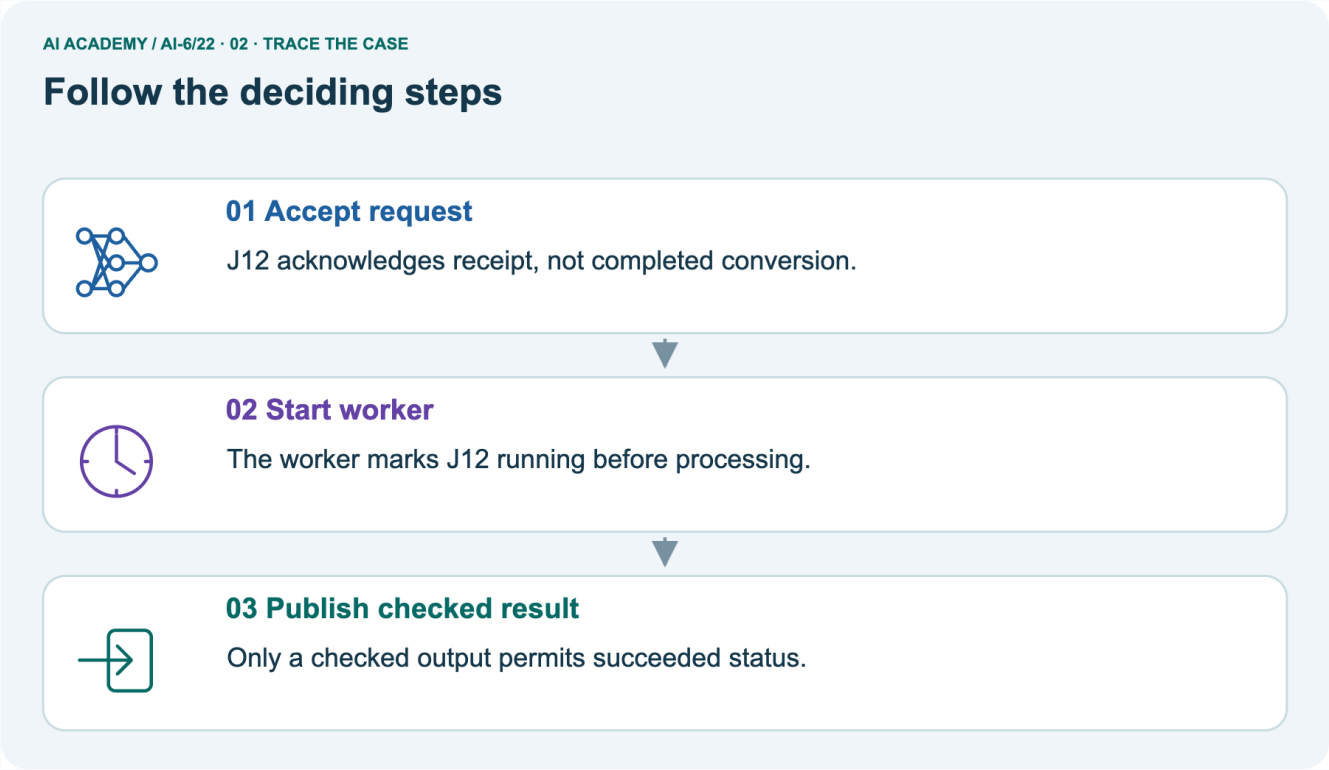

Follow the labelled picture with the matching explanation. Open the vector diagram

**Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission. Supplied facts: States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



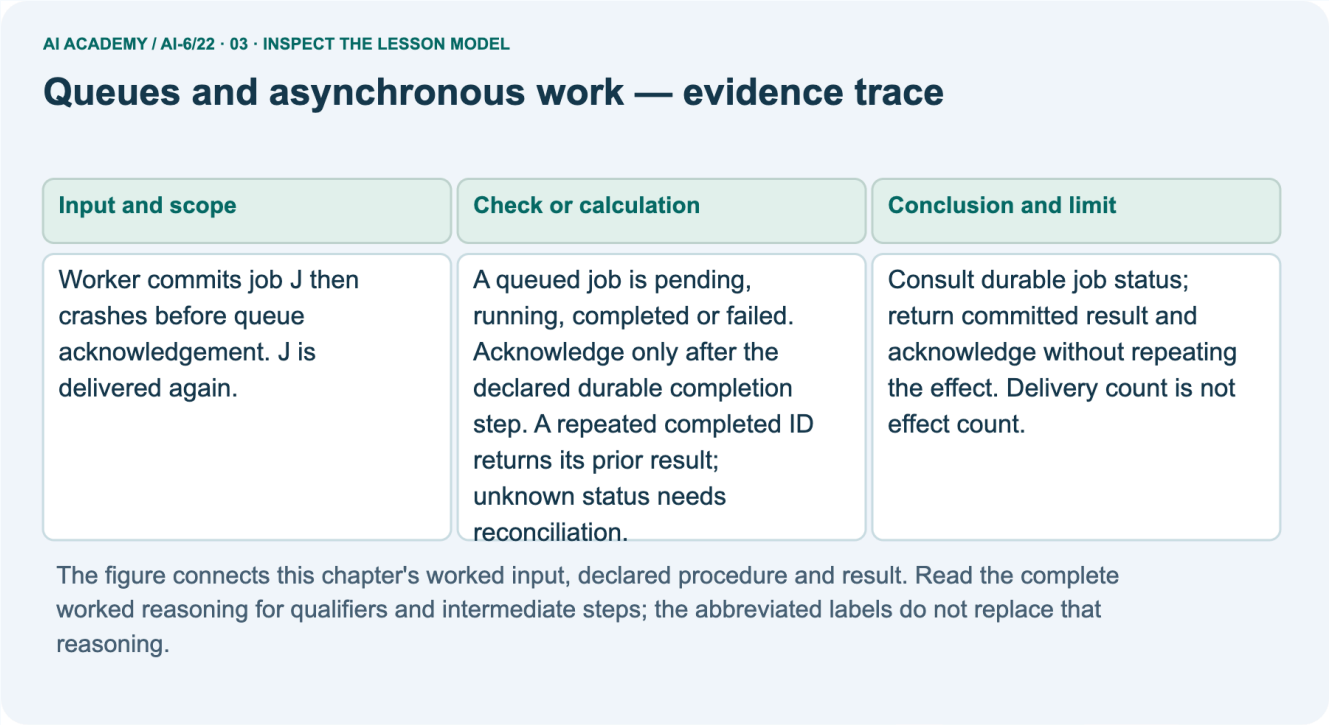

Follow the labelled picture with the matching explanation. Open the vector diagram

Accept request: J12 acknowledges receipt, not completed conversion. → Start worker: The worker marks J12 running before **process**ing. → Publish checked result: Only a checked **output** permits succeeded status.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

### 03. Queues and asynchronous work — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

**Queue**s and asynchronous work — **evidence** trace. Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. | A **queue**d job is pending, running, completed or failed.

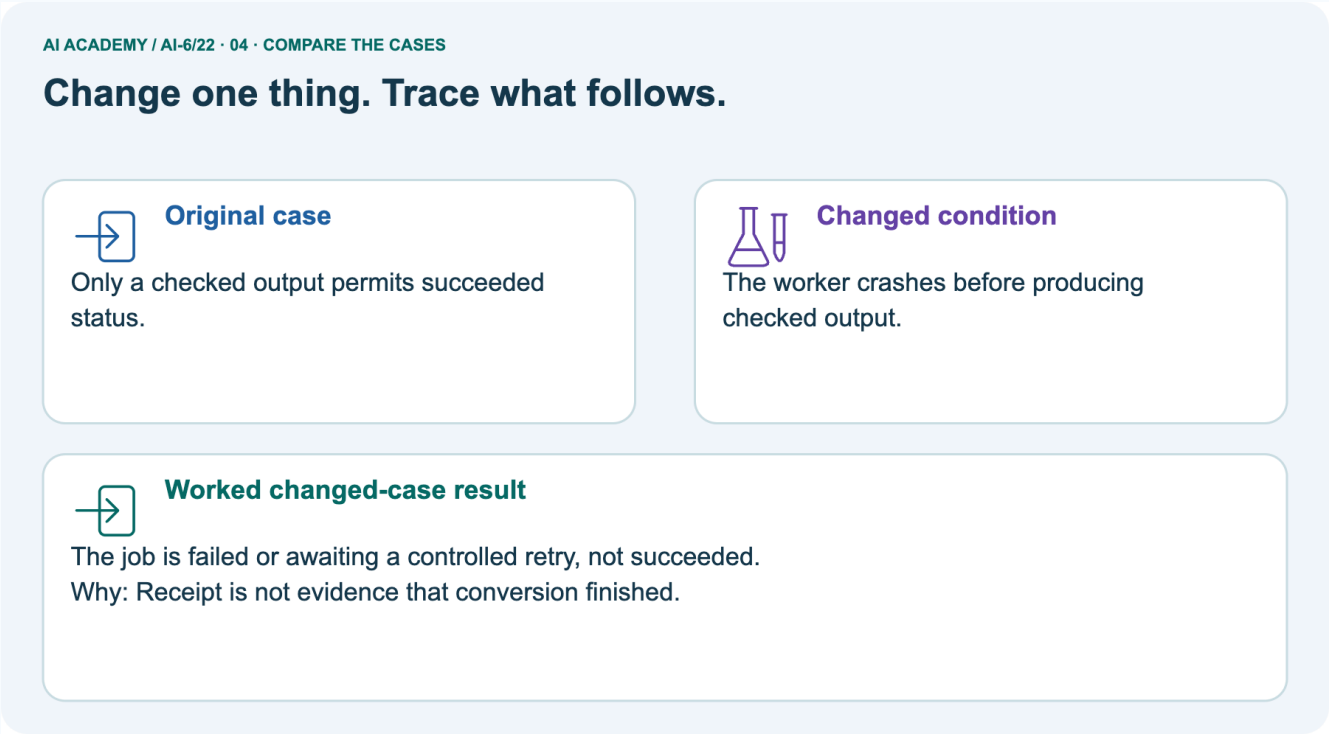
Acknowledge only after the declared durable completion step.

A repeated completed ID returns its prior result; unknown status needs reconciliation. | Consult durable job status; return committed result and acknowledge without repeating the effect. Delivery count is not effect count.

> Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

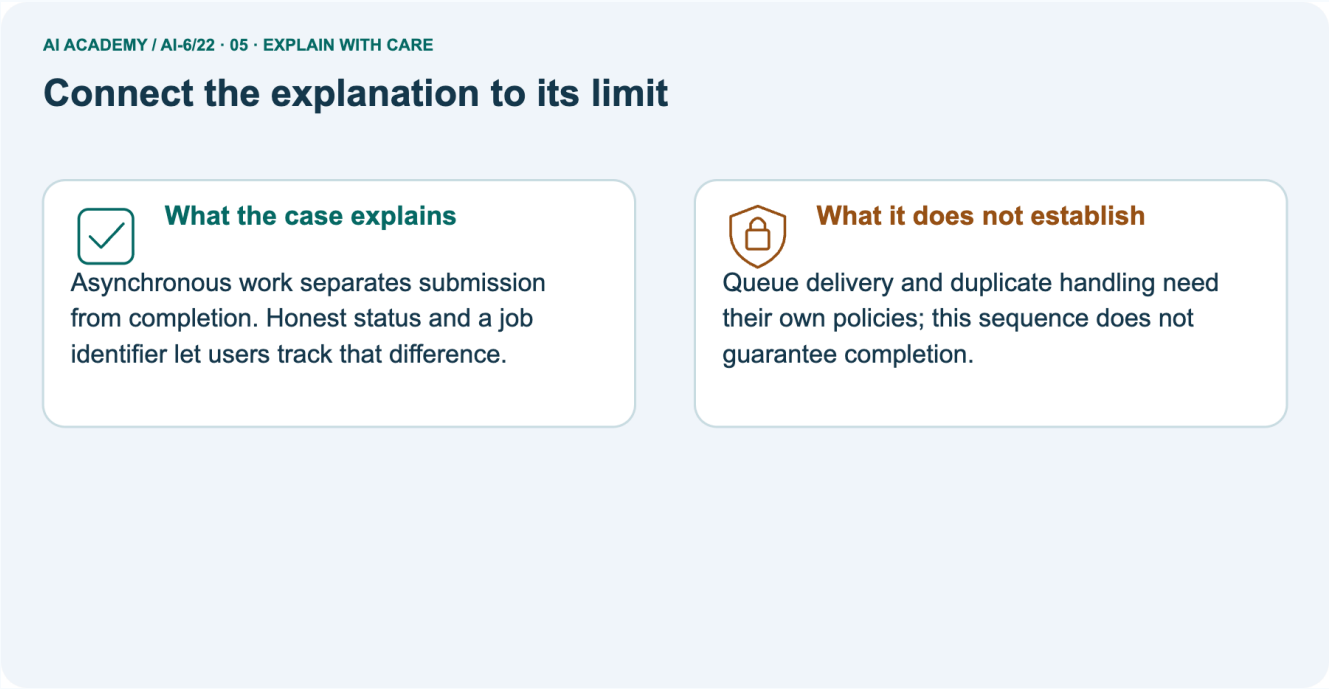
Original case: Only a checked **output** permits succeeded status. Changed condition: The worker crashes before producing checked **output**. Expected result: The job is failed or awaiting a controlled retry, not succeeded.

Reason: Receipt is not **evidence** that conversion finished.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

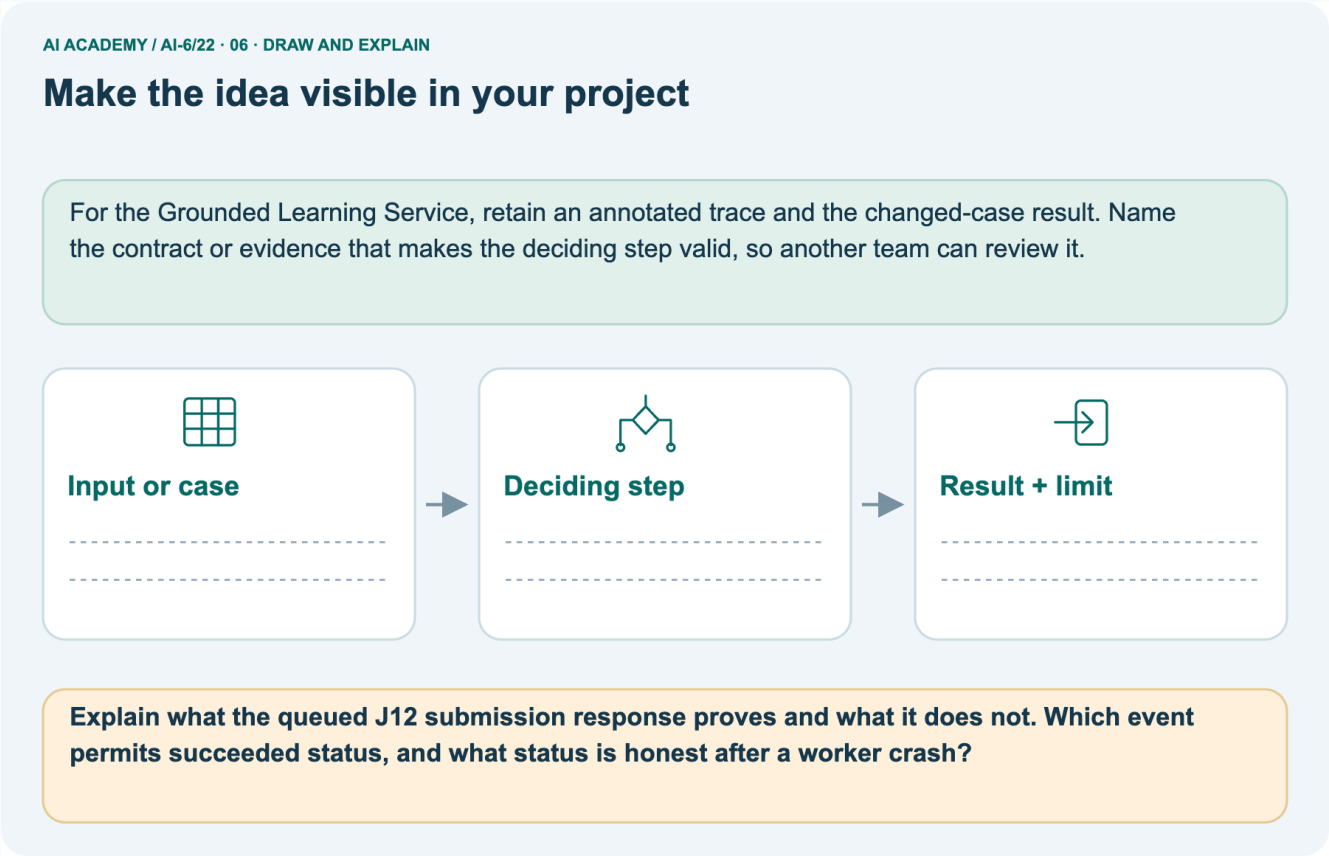
Asynchronous work separates submission from completion. Honest status and a job identifier let users track that difference.

Limit: **Queue** delivery and duplicate handling need their own policies; this sequence does not guarantee completion.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain what the **queue**d J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.